



#	Milestone	Generative AI Guidelines	Due	Mark Release
1	Project Definition & Team Formation	C: AI-Assisted Editing ¹	Sep. 23, Wed., AoE ²	Sep. 28, Mon.

This course is *project driven*. Students will work in teams to define, develop, and complete a project involving data modeling and representation over the course of one semester. The course project has two main objectives:

- Hands-on experience: Gain practical experience in analyzing and modeling real-world data, while developing skills in data representation, schema design, querying, implementation, evaluation, and teamwork.
- Research experience: Develop an understanding of the research *process*, from identifying a problem and reviewing existing approaches to designing, implementing, and evaluating different representations of data in the same domain.

Projects should aim to advance the state of the art, rather than simply reproduce an existing solution. This provides students with an opportunity to conduct meaningful project while completing the course requirements.

A data modeling project typically starts with a real-world domain and an information need. In this sense, a data modeling project may resemble a software or database project; however, there are important differences. A software project may primarily focus on implementing a functional system, while a data modeling project in this course focuses on understanding the structure and meaning of information and determining how that information can be represented, stored, and queried effectively for globally shared, accessible, and interoperable representations that can support Knowledge Graphs, the Giant Global Graph (GGG), and Linked Open Data³.

In this course, students will progressively explore different approaches to representing the same real-world domain. The project will begin with conceptual data modeling, including the identification of entities, attributes, and relationships and their representation using an Entity-Relationship Diagram (ERD). The conceptual model will then be transformed into a relational model (logical level of data modeling), including relational tables, keys, and constraints, and implemented and queried using SQL. The project will subsequently extend the representation toward semantic and knowledge representation, using RDF, RDFS, and OWL, and will explore the storage and querying of the resulting knowledge graph using a triple store and SPARQL:

Real-world domain → ERD → Relational Schema → SQL → RDF/OWL → Triple Store → SPARQL

The goal is not simply to implement these technologies independently. Rather, students should understand how different data models represent information and relationships, how one representation can be transformed into another, what information and semantics may be preserved or lost during such transformations, and how different representations affect the types of questions that can be answered.

Roughly, the data modeling project in this course consists of the following milestones (phases):

1. Project Definition & Team Formation (Proposal)
2. Background & Related Work (Literature Review)
3. Project Presentation: 3-Minute Talk (3MT)

¹ AI can be used to make improvements to the clarity or quality of student-created work to improve the final output, but no new content can be created using AI (e.g. language editing for lab reports). AI can be used, but your original work with no AI content must be provided in an appendix.

² <https://www.timeanddate.com/time/zones/aoe>

³ <https://www.dbpedia.org/>



4. Solution Design & Implementation (Data Modeling & Representation)
5. Evaluation (Experimentation)
6. Project Demonstration: 10-Minute Talk (10MT)

A project manual has been prepared to guide students through each milestone. Each milestone breaks the project into a series of smaller steps and deliverables. The current manual covers Milestone 1: Project Definition & Team Formation, also known as Proposal.

1.1. Define a Data Modeling Problem

The 1st step of the Proposal is to select and define a real-world data modeling problem. A data modeling problem arises when information about a real-world domain needs to be systematically represented, organized, stored, and queried. The goal is not simply to collect data, but to determine what information is important, how different pieces of information are related, and what kinds of questions should be answerable from the resulting data representation. Examples include universities, transportation systems, healthcare information, cultural heritage, scientific publications, e-commerce, social networks, sports, tourism, libraries, or other real-world domains.

For example, “*a municipal public safety and security*” domain contains information about:

- “*Security incidents and their types, such as theft, assault, vandalism, suspicious activity, and unauthorized access;*
- *People or agents involved in incidents, such as victims, subjects, witnesses, security officers, and other responders;*
- *Locations where incidents occur, including buildings, rooms, entrances, parking areas, and other campus locations;*
- *The date, time, and duration of security incidents;*
- *Security notifications, reports, CCTV observations, and other sources through which incidents are recorded;*
- *Security responses associated with incidents, such as officer dispatches, investigations, or emergency responses; and*
- *Relationships among incidents, such as incidents occurring at the same location, involving the same individuals, occurring within a particular time period, or being related to the same security investigation.”*

The project should identify a specific information-management problem within such a domain. For example, the problem may involve representing “*security incidents, people, locations, times, reports, and responses*” and supporting questions that require understanding the relationships among these entities and events.

A project should not simply be:

“Build a database for city security events.”

Instead, it should identify a specific data modeling problem, such as:

*“How can information about security incidents, people, locations, times, reports, and responses be represented in a globally accessible and interoperable form so that humans and **computational systems**, including software applications, autonomous agents, and large language models (LLMs), can consume, integrate, query, and reason over security events and their relationships?”*

The problem should be sufficiently rich to support the development and comparison of different data representations throughout the course. In particular, the selected problem should provide enough entities, attributes, and relationships to allow the team to develop a meaningful Entity-Relationship Diagram (ERD), transform it into a relational model, formulate SQL queries, and subsequently explore an RDF/OWL-based knowledge representation and SPARQL queries. The selected domain should also contain relationships that make semantic representation meaningful. For example, *“a security incident may involve multiple people, occur at a particular location and time, be reported by multiple sources, trigger one or more responses, and be related to other incidents.”* A project that consists almost entirely of *“independent incident”* records with few meaningful relationships is *not* appropriate for this course.

The project should therefore identify:

- The real-world domain and its scope;
- The important entities or concepts in the domain;
- The important attributes or properties of those entities;
- The relationships among entities, concepts, or events;
- The types of information that users need to retrieve; and
- The important questions that the data representation should be able to answer.

Important: Problem Definition vs. Proposed Solution

At this stage, students should focus on what information needs to be represented and what information needs to be obtained from it, rather than how the information will ultimately be represented or queried. Do not mix the problem definition with the solution. The problem definition should not prescribe the data model, database technology, query language, or implementation technique that will be used to solve the problem.

For example, avoid defining the problem as:

“We will design an ERD, convert it into relational tables, store the data in MySQL, and use SQL queries to retrieve information about security incidents. We will then represent security incidents using RDF and OWL and query them using SPARQL in Virtuoso triple store.”

This describes a proposed solution rather than the problem itself. These technologies belong to the later Milestone 4. Solution Design & Implementation. Instead, define the problem in terms of the required information and tasks:

“The task is to represent information about security incidents, the people and locations involved, the time and circumstances of incidents, their sources and responses, and the relationships among incidents, in a globally accessible, interoperable, and machine-understandable form so that large language models (LLMs) can retrieve and reason over the information and its relationships.”

A good problem definition should therefore answer:

- What information is given or available? What entities, concepts, or events are involved? What are the important attributes or properties of those entities, concepts, or events? How are they related in the real world?
- What information needs to be represented? What information may come from different sources or systems? What differences or ambiguities may exist in how different sources describe the same information?

Commented [A1]: Please don't simply copy/paste or follow this narration. Come up with your own way of defining task related to your project.

- What information should be shared, integrated, or reused across systems or domains? What information should be understandable not only to humans but also to computational systems?
- What questions should the resulting representation be able to answer for users or computational systems?
- What information should be connected to related information from other sources or domains? What information should be accessible and reusable beyond the original system or application?
- What requirements are needed for the representation to support future knowledge graphs, Linked Open Data, autonomous agents, or large language models (LLMs)?

The problem definition should be precise enough that another person could understand what the project is trying to represent and query without knowing the solution that the team intends to develop.

1.2. Establish Motivation and Significance

The 2nd step of the Proposal is to explain why the chosen domain is worth data modeling. We should not only ask “Can we model the data in this domain?” but also “Why does modeling data in this domain matter?”

You should identify a practical, scientific, or societal motivation for your project. In other words, ask:

- *So what, who cares?*
- *Why should we and others put effort into this?*
- *Who could benefit from it?*
- *What could become possible if the information were modelled or existing models were improved?*

For example, for our example domain “*a municipal public safety and security*”, you can explain how information about public-safety incidents may be produced independently by different organizations and systems, using different schemas, vocabularies, identifiers, and levels of detail, and why your project could be useful like:

“An interoperable representation of municipal public-safety and security information could potentially support:

- o *More effective integration of heterogeneous information sources, such as municipal incident reports, emergency-service records, security logs, and external threat-intelligence sources, even when these sources use different data structures or terminology [citations];*
- o *Improved semantic interoperability by providing shared concepts and formally defined relationships that allow independently developed datasets and information systems to exchange and interpret information consistently [citations];*
- o *More comprehensive investigation and analysis of incidents over time, by connecting related events and entities across multiple information sources [citations]; and*
- o *Greater reuse and accessibility of security knowledge, by representing information using shared, machine-interpretable concepts that can potentially be linked to external datasets, and ontologies [citations].”*

The motivation should go beyond simply asking whether the information can be stored in a database. A central question is whether different organizations and information systems can represent and understand related information in a common, machine-interpretable way.

Important: Claims about potential benefits should be supported by appropriate scientific or authoritative references/citations.

1.2.1. Motivating Example

At least one concrete example should demonstrate the potential benefit of your project. The example should make the motivation easy to understand by showing:

- A real-world information need or problem situation;
- The information and relationships that are relevant;
- The capability that an appropriate data representation could provide; and
- The potential benefit of that capability.

For example:

“Suppose a city experiences a security incident at a public facility. The municipality, police, emergency services, and external threat-intelligence providers independently record information about the incident, using different schemas, terminology, and identifiers. If these sources use a shared ontology and machine-readable semantic representation, their information could be linked, allowing a computer to recognize that events recorded by different systems concern the same location, facility, or incident.”

Commented [A2]: add exact city name, public facility name, time, date, very details of the event, and then how different systems are referring to the same event with different terminologies or standard.

The motivating example should explain “*why this way of data modeling matters*”, rather than explain how the database or knowledge graph will be implemented.

1.2.2. Gaps in Existing Solutions

Finally, you must explain why existing ways of representing or managing the information are not necessarily sufficient and, consequently, why your project is worth pursuing. There are several possibilities:

- Existing representations address a related problem but not your particular domain or setting;
- Existing representations capture entities or events but do not adequately capture important relationships among them;
- Existing approaches require substantial manual integration of heterogeneous information from logs, reports, databases, and sources;
- Existing representations are difficult to extend when new types of entities or relationships are introduced;
- Existing systems make assumptions that do not hold in your setting;
- Existing approaches make particular types of queries difficult or inefficient;
- Existing approaches do not adequately represent the semantics of relationships among entities; or
- Existing information is distributed across multiple sources and is difficult to integrate into a unified representation.

For example, after reviewing existing approaches, you might write:

“Although security information is commonly represented using structured databases, event logs, and specialized security data formats [citations], these representations may be designed for particular systems or purposes and may make integration across heterogeneous sources difficult. Furthermore, relationships among security incidents, events, affected systems, users, vulnerabilities, and attack techniques may not always be explicitly represented in a form that supports integrated semantic querying.”

Alternatively, you may discover that an existing representation already captures the required information but has a particular limitation:

“Although existing security-event representations can capture individual events and their attributes [citation], representing and querying relationships that span multiple heterogeneous sources may require extensive integration and complex queries. For example, connecting a phishing event to a compromised account, the systems subsequently accessed by that account, and vulnerabilities associated with those systems may require information from several sources. This motivates investigating alternative data representations that can explicitly capture the semantics and relationships among security entities and events.”

In summary, a strong proposal should provide convincing answers to two distinct questions for its motivation:

- Why is the data modeling problem important? → What practical, scientific, or societal benefit could result from better representing and connecting information?
- Why is further investigation necessary? → What limitation, gap, or opportunity remains in existing representations or solutions?

1.3. Author List

The 4th purpose of the Proposal is to create a team of a minimum of 2 and maximum of 3 students. Individual projects are highly discouraged and are permitted only in rare and exceptional cases. Team projects facilitate better workload distribution, earlier and more effective fault detection, and opportunities to develop collaboration and conflict-resolution skills. In a team project:

- The rationale for the members’ collaboration must be justified, such as through previous successful collaboration, shared experience in the research topic, or complementary expertise; and
- Each member’s contribution to every part of the research project must be clearly identified. Evaluation scores will be distributed among team members based on their individual contributions.

1.4. Submission Guidelines

1.4.1. Deliverables

- Submission must be written in English, in the current ACM two-column conference format in LaTeX. Overleaf⁴ templates are available from the ACM Website⁵. Use the *sigconf* proceedings template.
- The submission must be *exactly* 1 page in length, no more and no less, including author list, figures, tables, and references, with unlimited pages for C: AI-Assisted Editing appendix, and must be authored by all team members. We understand that fitting all the required content into exactly one page can be challenging. However, this is intentional: an important part of communication is learning how to retouch, refine, condense, and prioritize information without losing its essential meaning. We recommend initially writing a longer version (e.g., 3-5 pages) and then gradually reducing and retouching it until you can effectively communicate the key ideas within the required one-page limit.
- The submission must be provided as a single zip file named: 1_Proposal_firstname1_firstname2_[...].zip, e.g., 1_Proposal_Kovvy_Hosseini.zip, containing:
 - o The LaTeX source files and any required supplementary files;
 - o The compiled one-page PDF file.

A *sample* is attached to this manual and is also available [here](#)⁶ (1_Proposal.tex).

⁴ <https://www.overleaf.com/gallery/tagged/acm-official>

⁵ <https://www.acm.org/publications/proceedings-template>

⁶ <https://www.overleaf.com/read/vtgnmsqqfqv>

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1.4.2. Team and Individual Contributions

- Each team must maintain a Git-based repository (e.g., GitLab, GitHub, ...) ⁷ for the project. The repository must be associated with the individual accounts of all team members. The repository will be used to document the development of the project and to make individual contributions traceable. The repository should be used throughout the project rather than only immediately before a submission deadline. The instructor and/or marker may consider the repository's commit history, pull requests, issues, documentation, and other project records when assess individual contributions. The number of commits alone will not be used as a measure of contribution. The quality, significance, and nature of each member's contributions will also be considered.
- All team members are expected to make substantive contributions, and no member should be limited solely to coordination or project management.

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1.4.3. Research Problem Selection

You may not select the same research problem as another group unless you are members of the same team. Once you have identified a potential research problem, communicate your choice to the instructor and obtain approval before proceeding to the next step.

1.4.4. Marking Scheme

- 10% Abstract
- 20% Introduction
- 20% Motivation
- 10% Motivating example
- 20% Team members (author list, including affiliation) + collaboration justification
- 20% Style, format, repo, file naming convention, ...

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⁷ Web-based applications and cloud services may collect, use and/or disclose your personal information and store or access that information outside of Canada. Personal information includes, but not limited to, identifiable individual's information such as your name or your email address. At UVic, GitLab at gitlab.uvic.ca is highly recommended for coursework, research and development projects. For more information, see <https://www.uvic.ca/systems/services/pages/gitlab.php>. At your team's discretion, you may use GitHub to store information about your project, but we do not require nor recommend storing any personal identifying information. Also, we do not require public repo, but it should be accessible to instructors and markers for evaluation purposes.